JASON PHILLIP LU

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EDUCATION

University of Michigan *Expected May 2028*
Doctor of Philosophy, Civil Engineering GPA: 3.966/4.0
Advisor: Prof. Neda Masoud
Specialization in Next Generation Transportation Systems

University of Michigan *Expected May 2026*
Master of Science, Industrial and Operations Engineering GPA: 3.966/4.0

Georgia Institute of Technology *Dec. 2022*
Bachelor of Science, Industrial Engineering GPA: 3.93/4.0
Research Advisor: Prof. Pascal Van Hentenryck *Graduated with Highest Honors*
Minor in Scientific and Engineering Computing

RESEARCH INTERESTS

Methodologies optimization, machine learning, game theory, simulation, network modeling

Applications shared mobility, incentive mechanisms, connected and autonomous vehicles, logistics

PUBLICATIONS

Peer-Reviewed Journals

Lu, J., Masoud, N. (in press). Incentive Design for Mobility Systems: A Review, Insights, and Future Directions. *Transportation Research Part E: Logistics and Transportation Review.* http://doi.org/10.13140/RG.2.2.32670.60480

Lu, J., Trasatti, A., Guan, H., Dalmeijer, K., Van Hentenryck, P. (2024). The Impact of Congestion and Dedicated Lanes on On-Demand Multimodal Transit Systems [Special Issue: Multimodal Public Transport, Travel Behaviour and Social Equity]. *Travel Behaviour and Society.* Volume 36, 100772. https://doi.org/10.1016/j.tbs.2024.100772

Under Review

Lu, J., Santanam, T., Guan, H., Riley, C., Kim, M.S., Trasatti, A., Masoud, N., Van Hentenryck, P. (2025). The Impact of Shared Autonomous Vehicles in Microtransit Systems: A Case Study in Atlanta. *Under Review in Transportation Research Part C: Emerging Technologies.* https://doi.org/10.48550/arXiv.2601.21046

Lu, J., Xia, X., Bahrami, S., Masoud, N. (2026). Path-Based Incentives for Multimodal Networks under Multiclass Traffic Assignment. *Submitted to Transportation Research Part B: Methodological.* https://doi.org/10.13140/RG.2.2.15344.62722

Working Papers

Lu, J., Masoud, N. The Incentive Bundle Program: Behaviorally-Consistent Incentive Design for Multimodal Transportation. *In Preparation for Transportation Research Part B: Methodological.* https://doi.org/10.13140/RG.2.2.22027.14882

Lu, J., Masoud, N. Cooperation and Competition in Multimodal Transportation Incentive Programs: A Tri-level Game-Theoretic Optimization Approach. *Extended Abstract Accepted at the 2026 INFORMS Transportation Science and Logistics Conference.*

Lu, J., Chen, C., Liu, Y., Wu, Y.L., Kusari, A., Masoud, N. A Large-Scale Optimization and Simulation Framework for Sensor Configuration in Connected and Autonomous Vehicles. *Abstract Submitted to the 2026 INFORMS Annual Meeting.*

PRESENTATIONS

Invited Talks

A Large-Scale Optimization and Simulation Framework for Sensor Configuration in Connected and Autonomous Vehicles. *Next Generation Transportation Systems Seminar Series*, Department of Civil and Environmental Engineering, University of Michigan, March 19, 2026.

The Incentive Bundle Program to Enhance Multimodal Transport. *Next Generation Transportation Systems Seminar Series*, Department of Civil and Environmental Engineering, University of Michigan, February 27, 2025.

Conference Presentations

Lu, J., Masoud, N. Cooperation and Competition in Multimodal Transportation Incentive Programs: A Tri-level Game-Theoretic Optimization Approach. *2026 INFORMS Transportation Science and Logistics Conference*, Boston, Massachusetts, July 26-29, 2026.

Lu, J., Chen, C., Liu, Y., Wu, Y.L., Kusari, A., Masoud, N. A Large-Scale Optimization and Simulation Framework for Sensor Configuration in Connected and Autonomous Vehicles. *2026 CCAT Global Symposium on Mobility Innovation*, Ann Arbor, Michigan, April 17, 2026.

Lu, J., Xia, X., Bahrami, S., Masoud, N. Can Incentives Reshape How We Move? A Multimodal Traffic Assignment Perspective. *The 105th Transportation Research Board Annual Meeting*, Washington, D.C., January 11-15, 2026.

Lu, J., Masoud, N. A Game Between Ridehailing Operator and Government Under the Incentive Bundle Program. *2025 INFORMS Annual Meeting*, Atlanta, Georgia, October 26-29, 2025.

Lu, J., Masoud, N. The Incentive Bundle Program to Enhance Multimodal Transportation. *The 7th Bridging Transportation Researchers Conference*, Virtual, August 6-7, 2025.

Masoud, S., Masoud, N., Alinezhad, E., **Lu, J.**, Rapp, S., Rimanelli, J. Adaptive Contested Logistics: A Dynamic Data-Driven Application System Framework with Dynamic Bayesian Belief Networks. *2025 ARC Annual Program Review*, Ann Arbor, Michigan, June 17-18, 2025.

Masoud, N., **Lu, J.**, Chen, C., Liu, Y., Wu, Y., Kusari, A. Balancing Safety, Cybersecurity, and Mobility: Quantifying the Impact of Sensor Redundancy in Connected and Autonomous Vehicles. *2025 CCAT Global Symposium on Mobility Innovation*, Ann Arbor, Michigan, March 28, 2025.

Lu, J., Masoud, N. Designing Incentive Bundles for Multimodal Transportation Systems. *2024 INFORMS Annual Meeting*, Seattle, Washington, October 20-23, 2024.

Akhlaghi, V.E., Guan, H., **Lu, J.**, Van Hentenryck, P. Optimizing Truck Fleet Scheduling for Fuel Deliveries. *2023 INFORMS Annual Meeting*, Phoenix, Arizona, October 15-18, 2023.

HONORS AND AWARDS

Western Track Best Paper Award *Sep. 2025*
Bridging Transportation Researchers Conference

Rackham Conference Travel Grant *Aug. 2024, Aug. 2025*
Rackham Graduate School, University of Michigan

Seth Bonder Fellowship *Jun. 2022, Jun. 2023*
H. Milton Stewart School of Industrial and Systems Engineering, Georgia Institute of Technology

Stewart Topper Fellowship (Declined) *Feb. 2023*
H. Milton Stewart School of Industrial and Systems Engineering, Georgia Institute of Technology

Senior Design Capstone Finalist *May 2022*
H. Milton Stewart School of Industrial and Systems Engineering, Georgia Institute of Technology

SERVICE - INTERNAL

Michigan Transportation Student Organization
Vice President *May 2024 - Apr. 2025*

The Seth Bonder Camp in Computational and Data Science for Engineering
Instructor *Jun. 2022 - Jul. 2023*

SERVICE - EXTERNAL

Reviewer for Academic Journals and Conferences

- Transportation Research Board
- Bridging Transportation Researchers

Conference Session Chair

- Advancements in Urban Transit Systems, *2024 INFORMS Annual Meeting*, Seattle, Washington, October 20-23, 2024.

ADVISING

Masters Students

Yuyang Liu, University of Michigan
M.S. Electrical Engineering (Expected 2026)
Placement: PhD Student, University of Michigan

Undergraduate Students

Yi Ling Wu, University of Michigan
B.S. Computer Engineering (2025)
Placement: PhD Student, Texas A&M University
- Aviles-Johnson Doctoral Fellowship

TEACHING

Graduate Student Instructor **University of Michigan**

CEE 850 - Next Generation Transportation Systems Seminar *Spring 2024*
Instructor: Prof. Neda Masoud

Guest Lecturer **University of Michigan**

Gurobi *Sep. 2024, Sep. 2025*
CEE 553 - Infrastructure Systems Optimization

Overview of Next Generation Transportation Systems Program *Mar. 2025*
CEE 200 - Introduction to Civil and Environmental Engineering

Preparing for the Civil and Environmental Engineering Career Fair *Sep. 2024*
CEE 200 - Introduction to Civil and Environmental Engineering

Head Teaching Assistant **Georgia Institute of Technology**

ISYE 4134 - Constraint Programming *Spring 2023*
Instructors: Mr. Tejas Santanam, Prof. Pascal Van Hentenryck

Undergraduate Teaching Assistant **Georgia Institute of Technology**

ISYE 3044 - Simulation Analysis and Design *Summer 2022*
Instructor: Prof. Seong-Hee Kim

ISYE 4034 - Decision and Data Analytics *Spring 2022*
Instructor: Prof. Jye-Chyi Lu

ISYE 3770 - Statistics and Applications *Spring 2021*
Instructor: Dr. Tuba Ketenci

INDUSTRY EXPERIENCE

Convoy
Process Improvement Intern *Jan. 2022 - May 2022*

Yokogawa
Industrial Engineering Intern *May 2021 - Dec. 2021*
Industrial Engineering Intern *May 2020 - Jul. 2020*